

1
2 Personalized AI assistants could either promote or hinder the
3 emergence of collective intelligence [**SWP: *The potential***
4 ***impact of personalized AI assistants on collective***
5 ***intelligence?***]

6
7 Jaeseung Kim¹ Sang Woo Park^{1,2,*}

8 **1** School of Biological Sciences, Seoul National University, Seoul, Korea

9 **2** Institute for Data Innovation in Science, Seoul National University, Seoul, Korea

10 *Corresponding author: sangwoopark@snu.ac.kr

11 **Abstract**

12 Advances in artificial intelligence (AI) have enabled the wide use of personalized
13 assistance for cognitive tasks, which enhanced individual performance on various
14 fields. However, the impact of AI assistance on collective intelligence remains poorly
15 understood. AI assistance may undermine the emergence of collective intelligence by
16 reducing the individual diversity or introducing structured biases into a population.
17 Here, we develop a human-AI collective prediction model mediated by social learning
18 based on incentive structure by extending the existing works on collective intelligence.
19 Our model uncovers that emergence of collective intelligence is depend on how human
20 and AI interact. When AI works as if it is omniscient and gives assistance regardless
21 of users' interests, collective intelligence will be promoted. However, when AI works
22 as a chat-bot and gives assistance only based on users' interests, collective intelligence
23 could be hindered due to diversity loss and overconfidence on AI. We showed that,
24 to promote collective intelligence, penalizing or incentivizing AI usage is not reliable.
25 Instead, elaborating more diversity-aware incentives could be the key to promote and
26 maintain the collective intelligence.

Introduction

Collective intelligence (CI) describes a phenomenon in which a group’s ability to solve problems, make decisions, or adapt to changing environments exceeds each member’s ability [1, 2, 3, 4, 5]. This phenomenon, also referred to as wisdom of crowds, has thus served as a central mechanism underlying the success of diverse organizations, ranging from animal groups [6, 7, 8] and human societies [2, 9] to machine-learning systems [10, 11]. Understanding the conditions that promote and maintain CI therefore remains a fundamental question across social sciences [12], evolutionary biology [13], and computer science [14]. Addressing this question has become particularly timely as artificial intelligence (AI) has begun to reshape how individuals acquire information, make decisions, and contribute to collective outcomes [15].

The recent emergence of generative AIs and large language models (LLMs) has changed the way individuals engage in various cognitive tasks, ranging from simple everyday tasks to more complex tasks, including scientific discovery [16] and creative writing [17]. In particular, while the use of generative AI models, such as ChatGPT, has been rapidly increasing [18], the potential long-term impact of personal AI assistants on human society, especially the CI, remains uncertain. Some argue that AI can promote CI by enhancing individual expertise [19], and opinion aggregation [20]. Conversely, empirical evidence indicate that the use of AIs can limit the diversity of opinions [17, 21, 22], a key requirement for the emergence of CI [23, 24, 25].

Our ability to predict the long-term impact of personal AI assistants on human CI is further hampered by the apparent discrepancy in the incentives that shape AI uses. On one hand, many industrial sectors are pushing towards increased AI usages, where the number of “tokens” used is often perceived as a proxy for productivity [26, 27]. On the other hand, many academic institutions appear to discourage, and even penalize, AI uses, reflecting concerns about plagiarism, misinformation, and limited independence [28, 29]. Such discrepancy raises a question about whether directly incentivizing or penalizing AI usage will benefit or harm human CI. These observations further highlight the importance of identifying a robust incentive structure that can promote CI in the presence of personal AI assistants. Such incentive structure should seek to retain the benefits of AIs and prevent potential risk, such as preserving the diversity of individual opinions.

To address these questions, we extend previous approaches for modeling CI, where each individual makes independent predictions for the outcome from partial observations of underlying causal factors [12, 13, 30]. In particular, we allow each individual to independently consult with personal AI assistants and revise their predictions accordingly. The resulting model allows us to investigate the robustness of human CIs under different assumptions about the AI model’s ability to make predictions as well as incentive structures that are associated with the prediction task. In doing so, we assess the impact of directly incentivizing or penalizing AI uses on CI and further suggest a balanced incentive structure that can maintain CI in the presence of AI.

68 **Main**

69 **Model**

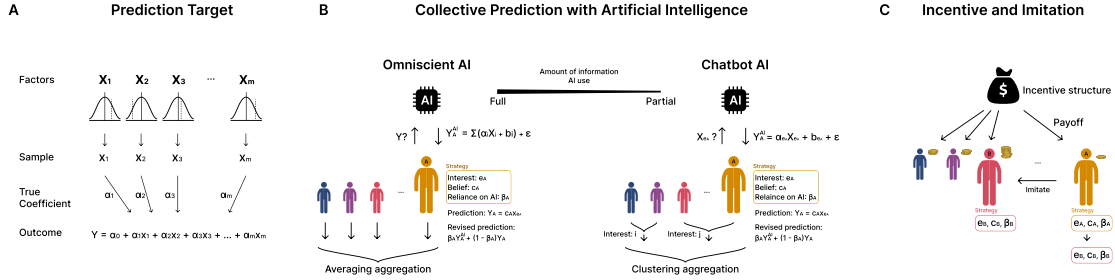


Figure 1: **A schematic diagram of the model for individual and collective prediction in the presence of personal AI assistants.** (A) The prediction target is modeled as a linear combination of causal factors. (B) Each player is characterized by three strategies: interest, belief, and reliance on AI. Players make predictions based on their interests and beliefs and subsequently revise them with AI assistance. We consider two types of AI, Chatbot AI and Omniscient AI, which differ in the amount of information available for generating predictions. (C) Players receive payoffs according to the incentive structure, and the population evolves through imitation of peers with higher payoffs.

70 We begin by extending previously established models of CI, in which a group of
 71 players participates in a prediction task (Figure 1): each player makes an individual
 72 prediction based on partial observations of the underlying causal factors, and the
 73 group forms a collective prediction by aggregating these individual predictions [30,
 74 12, 13]. Specifically, the prediction target (Y) is modeled as a linear combination of
 75 M causal factors x_i , with fixed coefficients α_i : $Y = \sum_{i=1}^M \alpha_i x_i + \alpha_0$, where each factor
 76 is drawn from independent normal distribution (Figure 1A). Then, each individual
 77 A observes a single factor of their interest x_{e_A} and makes a prediction based on
 78 their own belief c_A : $Y_A = c_A x_{e_A}$ (Figure 1B). We assume each individual can then
 79 independently consult their AI assistants and revise their prediction based on their
 80 reliance on AI β_A : $Y_A^{re} = \beta_A Y_A^{AI} + (1 - \beta_A) Y_A$ (Figure 1B). The revised predictions are
 81 then aggregated using two alternative aggregation rules, averaging and clustering, to
 82 form collective prediction.

83 To model AI, we assume that AI has already been trained on similar tasks and is
 84 able to make predictions based on partial or full observations of causal factors (Figure
 85 1B). Every time a player consults AI, it will make a new prediction Y_A^{AI} based on
 86 its internal parameters, which can include factor-specific bias (b_i) and stochastic
 87 error ϵ . Here, we consider two extreme types of AI response, which differ in the
 88 amount of information AI uses when making predictions. In one extreme, the AI
 89 can observe all causal factors and use all information to make predictions, regardless

of the user’s interest: $Y_A^{AI} = \sum_{i=0}^M (\alpha_i x_i + b_i) + \epsilon$. In the other extreme, AI works like a chatbot, where the AI predicts the outcome based only on the user’s interest: $Y_A^{AI} = \alpha_{e_A} x_{e_A} + b_{e_A} + \epsilon$. Hereafter, we refer to these two types as ‘Omniscient AI’ and ‘Chatbot AI’, respectively.

[JS: We assume that appropriate aggregation rule for collective prediction depends on the type of AI. With Omniscient AI, each revised prediction contains information of all causal factors and therefore represents an estimate of the full prediction target. With Chatbot AI, each revised prediction remains an estimate of the contribution of a single factor to the target. Based on this distinction, we use averaging aggregation for Omniscient AI, in which revised individual predictions are simply averaged to form the collective prediction. For Chatbot AI, we use clustering aggregation, in which revised predictions are first averaged among individuals interested in the same factor, and then summed across factors.]

Each generation consists of multiple round of predictions, and each player is rewarded a payoff according to the incentive structure (Figure 1C). For simplicity, we assume that a sufficiently large number of prediction rounds occurs within each generation, allowing us to use the expected payoff value rather than computing the payoffs for every round. We consider two incentive structures from previous work, “feedback” and “niche-expert”, which have been shown to promote near-perfect collective accuracy in the absence of AI [13]. The feedback structure gives higher payoff to individuals who can push the collective prediction toward the true target, regardless of their individual predictive accuracy: *[JS: $\pi_A = Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}})$]*. In contrast, the niche-expert structure gives higher payoff to individuals who have both better predictive accuracy and a minority interest: *[JS: $\pi_A = -\rho_{e_A}(Y_A^{\text{re}} - Y)^2$]*. *[JS: Here, ρ_{e_A} denotes the proportion of players interested in the same factor, thereby allowing the niche-expert structure to preserve interest diversity.]* We note collective intelligence fails to emerge under the combination of averaging aggregation with niche-expert structure in the absence of AI [13].

Through time, the group evolves by allowing each individual to imitate peers who received higher payoffs (Figure 1C). Specifically, after each generation, two individuals are randomly sampled, and one imitates all three strategies (the factor of interest, factor belief, and reliance on AI) of the other with a probability determined by the difference in their expected payoffs *[JS: (see the Simulations section in Methods)]*. Individual predictions are aggregated into a collective prediction $\hat{Y}$ based on the aggregation rules. Collective accuracy is defined as the reduction in prediction error achieved by the collective prediction relative to a baseline prediction α_0 : *[JS: $1 - \frac{\mathbb{E}[(Y - \hat{Y}^{\text{re}})^2]}{\mathbb{E}[(Y - \alpha_0)^2]}$]*. Hereafter, we say that CI emerges when the collective accuracy reaches a near perfect level.

This model now allows us to explore the impact of AI on CI depending on the mode of AI and the incentive structure. Here, we begin by comparing the emergence of CI with and without AI and subsequently investigate a how different incentive structures affect the emergence of CI in the presence of AI.

132 AI can either promote or prevent the emergence of CI

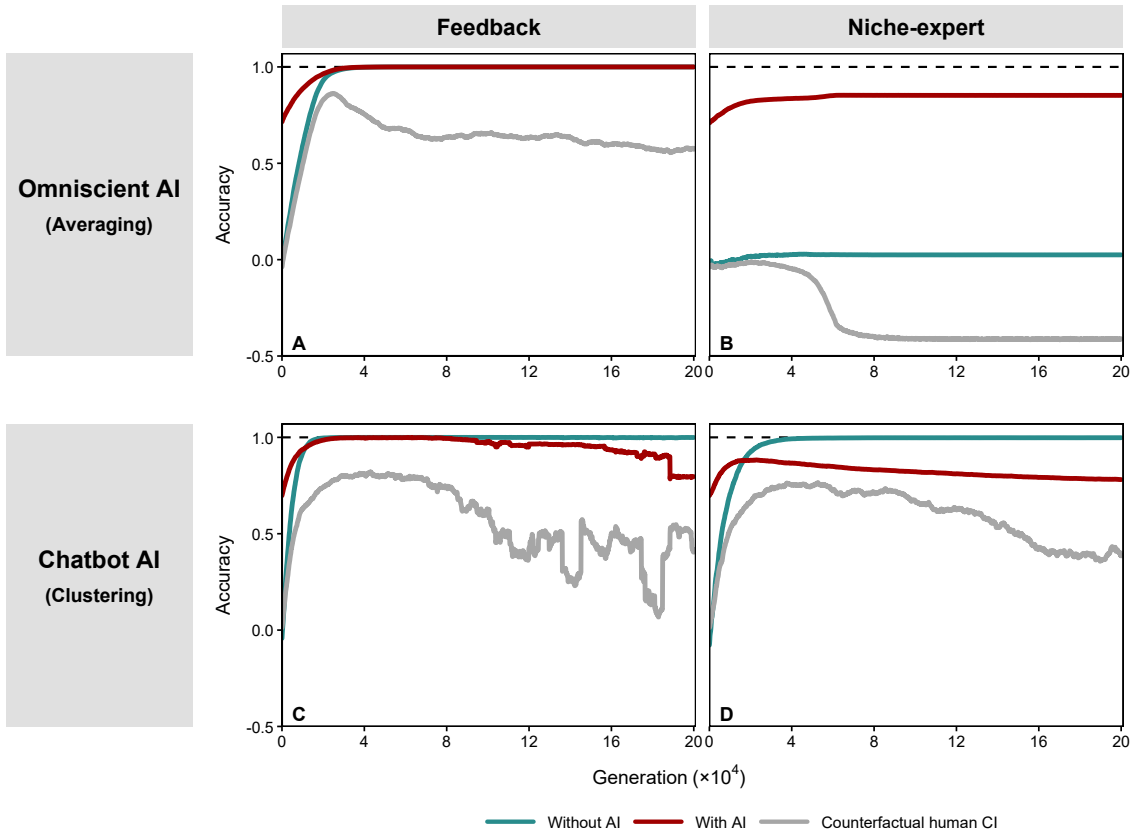


Figure 2: **Comparison of collective accuracy with and without AI.** Trajectories of collective accuracy when the group is assisted by the Omniscient AI (A,B) and Chatbot AI (C,D). *[SWP: Maybe we should also name the averaging rule below AI...]* Here, feedback (A,C) and niche-expert (B,D) incentive structures are considered. Green lines represent trajectories of collective accuracy without AI assistance. Red lines represent trajectories of collective accuracy with AI assistance. Gray lines represent trajectories of counterfactual collective accuracy, computed from players' unrevised predictions, indicating the group's predictive accuracy if AI assistance were unavailable at the given generation. Prediction target parameters: $m = 50$, $s = 50$, $\mu = 0$, $\alpha_i \sim U(-5, 5)$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10,000$, $c_A \sim \mathcal{N}(0, 100^2)$ for (A, B), $c_A \sim \mathcal{N}(0, 5^2)$ for (C, D), and $\beta_A \sim U(0, 1)$. AI parameters: $b_i = 0.36$ and $\tau = 0.3$.

133 To understand the impact of AI, we first compare collective dynamics with and
 134 without AI. First, consistent with [13], simple imitation rules and incentive struc-
 135 tures allow collective intelligence to emerge in the absence of AI (Fig. 2A,C,D). One
 136 exception is the use of simple averaging with niche-expert incentive structures, un-
 137 der which the collective accuracy of the population remains near zero. *[JS: This is*

138 *because enhancing individual accuracy causes individuals to predict close to the true*
139 *coefficient of their interest, but simply averaging these predictions does not neces-*
140 *sarily produce an accurate estimate of the full prediction target.][SWP: because...]*
141 (Fig. 2B) [SWP: maybe need a supp figure].

142 We find that AI usage can have heterogeneous outcomes depending on its type.
143 For example, the use of Omniscient AI can either preserve CI under the feedback
144 incentive structure (Fig. 2A) or promote higher collective accuracy under the niche-
145 expert incentive structure (Fig. 2B). These results are robust across different levels
146 of AI accuracy and can be shown analytically by assuming a sufficiently large group
147 of players (Fig. S1; see SI Evolutionary dynamics section).

148 In contrast, the use of Chatbot AI prevents the emergence of CI, thereby resulting
149 in lower levels of collective accuracy (Fig. 2C, D). Under the feedback incentive
150 structure, the use of Chatbot AI leads to a loss of diversity in the population's
151 interest, making it impossible to account for effect of some causal factors (Fig. S1,
152 S2). This causes an increase in collective variance and stochastic decrease in collective
153 accuracy (Fig. 1, S1)[SWP: Supp]. We note that the loss of interest diversity may
154 arise even in the absence of AI in finite populations (Fig. S3), but the use of Chatbot
155 AI makes the group more vulnerable to such loss (Fig. S3, S4). Under the niche-
156 expert incentive structure, the population becomes overly reliant on AI, causing the
157 collective prediction to inherit the AI's bias (Fig. S2). This leads to an increase
158 in collective bias and a decrease in collective accuracy (Fig. 2, S2) [SWP: Supp].
159 Interestingly, we find that the niche-expert structure can prevent the use of AI under
160 specific assumptions about the relationship between the true coefficient and AI bias
161 (Fig. S1). In this case, the population completely excludes AI assistance, allowing
162 the CI to emerge eventually. However, this phenomenon can substantially delay the
163 emergence of CI (Fig. S5) [SWP: Supp].

164 To investigate whether the group can retain its predictive ability when AI assis-
165 tance suddenly becomes unavailable at any given time, we quantify counterfactual
166 collective accuracy based on unrevised predictions. Across all scenarios, we find that
167 the counterfactual collective accuracy is lower than the collective accuracy of the
168 corresponding population.

169 In summary, an omniscient AI can promote CI, but AI operating with limited in-
170 formation may prevent the emergence CI. Moreover, reliance on AI can limit inherent
171 ability of the population to make accurate predictions on their own, raising concerns
172 about future AI usage. These findings further underscore the need for incentive
173 structures that can mitigate the potential risks associated with AI assistance.

174 **Directly incentivizing or penalizing AI use can additionally** 175 **harm CI**

176 Given the strong influence of AI on collective outcomes, a straightforward interven-
177 tion is to directly incentivize or penalize the use of AI. [SWP: maybe a sentence
178 *about how this reflects real world scenarios.]] To investigate whether such direct*

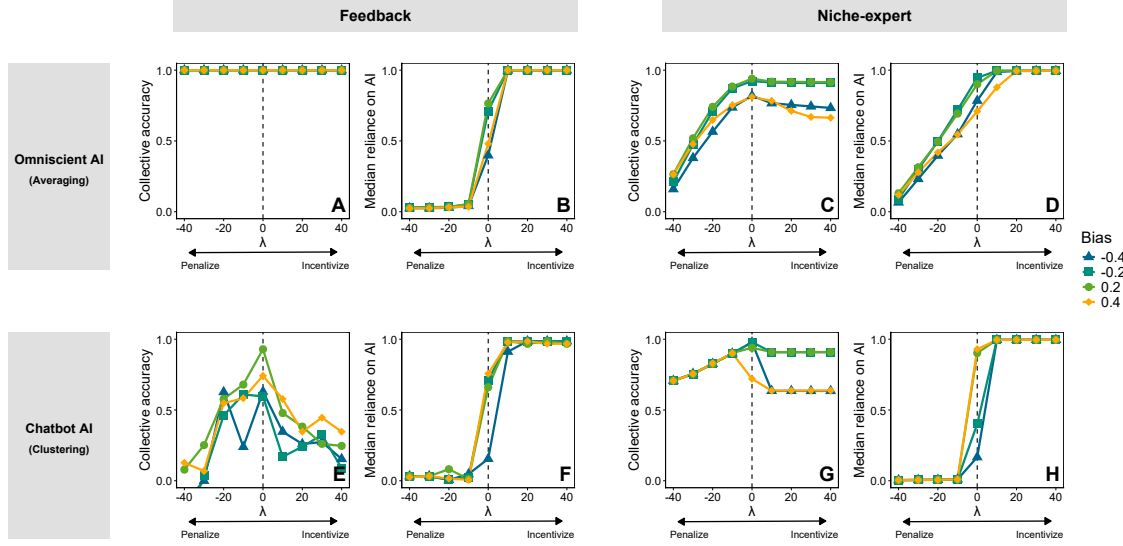


Figure 3: **Collective accuracy under the influence of direct intervention on AI use.** Collective accuracy and median reliance on AI under varying strengths of intervention (λ) across different assumptions about AI types (omniscient AI, A–D and chatbot AI, E–H) and incentive structures (Feedback, A,B,E,F, and Niche expert, C,D,G,H). Positive λ represents an incentive for AI use, whereas negative λ represents a penalty for AI use. In each panel, the vertical dashed line indicates the original incentive structure ($\lambda = 0$). Four AI bias levels are shown: $b_i = \pm 0.2$, representing higher-accuracy AI, and $b_i = \pm 0.4$, representing lower-accuracy AI. Collective accuracy (A,C,E,G) and median reliance on AI (B,D,F,H) are averaged over generations 190,000–200,000. Prediction target parameters: $m = 50$, $s = 50$, $\mu = 0$, $\alpha_i \sim U(-5, 5)$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10,000$, $c_A \sim \mathcal{N}(0, 100^2)$ for (A–D), $c_A \sim \mathcal{N}(0, 5^2)$ for (E–H), and $\beta_A \sim U(0, 1)$. AI parameter: $\tau = 0.3$.

incentives or penalties can help the emergence of CI, we modify the two incentive structures by adding a payoff term proportional to individual reliance on AI:

$$\text{Feedback/Niche-expert} + \lambda\beta_A \quad (1)$$

where λ represent the strength of intervention ($\lambda > 0$) or penalty ($\lambda < 0$). [SWP: maybe we need to call incentive pi earlier on...] [SWP: also good to define beta again here as a reminder.]

Directly incentivizing or penalizing AI use can further reduce the collective accuracy (Fig. 3C, E, G), except when Omniscient AI is used under the feedback structure; in the latter case, the population retains collective intelligence regardless of the intervention strength (Fig. 3A). The reduction in collective accuracy arises because incentivizing or penalizing AI usage causes individuals to imitate others based on their AI usage rather than their predictive accuracy. Thus, stronger interventions

190 can push the group to always ($\beta = 1$) or never ($\beta = 0$) rely on AI (Fig. 3B,D, F, H)
 191 but these scenarios do not necessarily lead to better outcomes (Fig. 3D, F, H). When
 192 AI bias is relatively large, a weak penalty can lead to higher collective accuracy, but
 193 this represents a narrow parameter regime (Figure 3G).

194 Overall, these results indicate that simply incentivizing or penalizing the AI, in-
 195 dependent of individuals’ performance, use may exacerbate rather than resolve the
 196 risk of AI on CI. Thus, we also considered an incentive structure where interven-
 197 tions for AI usage affects the payoffs multiplicatively, thus giving higher payoffs to
 198 individuals who rely more (or less) AI and makes better prediction. [SWP: Supp.]

199 **Balanced incentives mitigate AI-induced failures of CI**

200 Neither the original incentive structures nor simple interventions targeting AI use are
 201 sufficient to promote CI. What incentive structure, then, would work in the presence
 202 of AI? Here, we propose a “balanced” incentive structure, modeled as the average of
 203 the feedback and niche expert incentives.

204 The core idea behind the balanced structure is to maintain interest diversity and
 205 prevent collective bias. On one hand, the feedback incentive structure is prone to
 206 the loss of interest diversity but can prevent collective bias by incentivizing players
 207 whose predictions lie opposite to the collective prediction. On the other hand, the
 208 niche-expert structure can maintain interest diversity because it directly incentivizes
 209 minority interests. However the niche-expert structure cannot prevent collective bias
 210 because, within the same factor of interest, individuals are rewarded only according
 211 to their individual predictive performance, which in turn homogenize the predictions.

212 Therefore, when both incentive structures are combined, the population achieves
 213 CI by maintaining both interest diversity and limited collective bias (Fig. 4, S3).
 214 Compare to the feedback and niche-expert incentive structures, individuals are both
 215 evenly distributed through factors (Fig. 4C–E), and their beliefs lie closer to true
 216 values (Fig. 4F–H). Moreover, the balanced incentive structure allows for higher lev-
 217 els of counterfactual collective accuracy compared to pure feedback and niche-expert
 218 scenarios. These patterns are driven by the emergent property of the balanced struc-
 219 ture, in which group increases its reliance on AI as the AI becomes more accurate,
 220 demonstrating appropriately calibrated AI use.

221 [SWP: *may need some robustness check rather than a case study...*]

222 **Discussion**

223 The ongoing race towards the development of “better” AIs continues to pose ques-
 224 tions about the societal benefits and risks of AI uses. Here, we used mathematical
 225 models of evolutionary dynamics to investigate how AI uses and incentive structures
 226 jointly affect the emergence of CI, which is defined as the population’s ability to make
 227 accurate predictions. We find that an “omniscient” AI model, which incorporates in-
 228 formation about all relevant causal factors, can improve collective accuracy, whereas

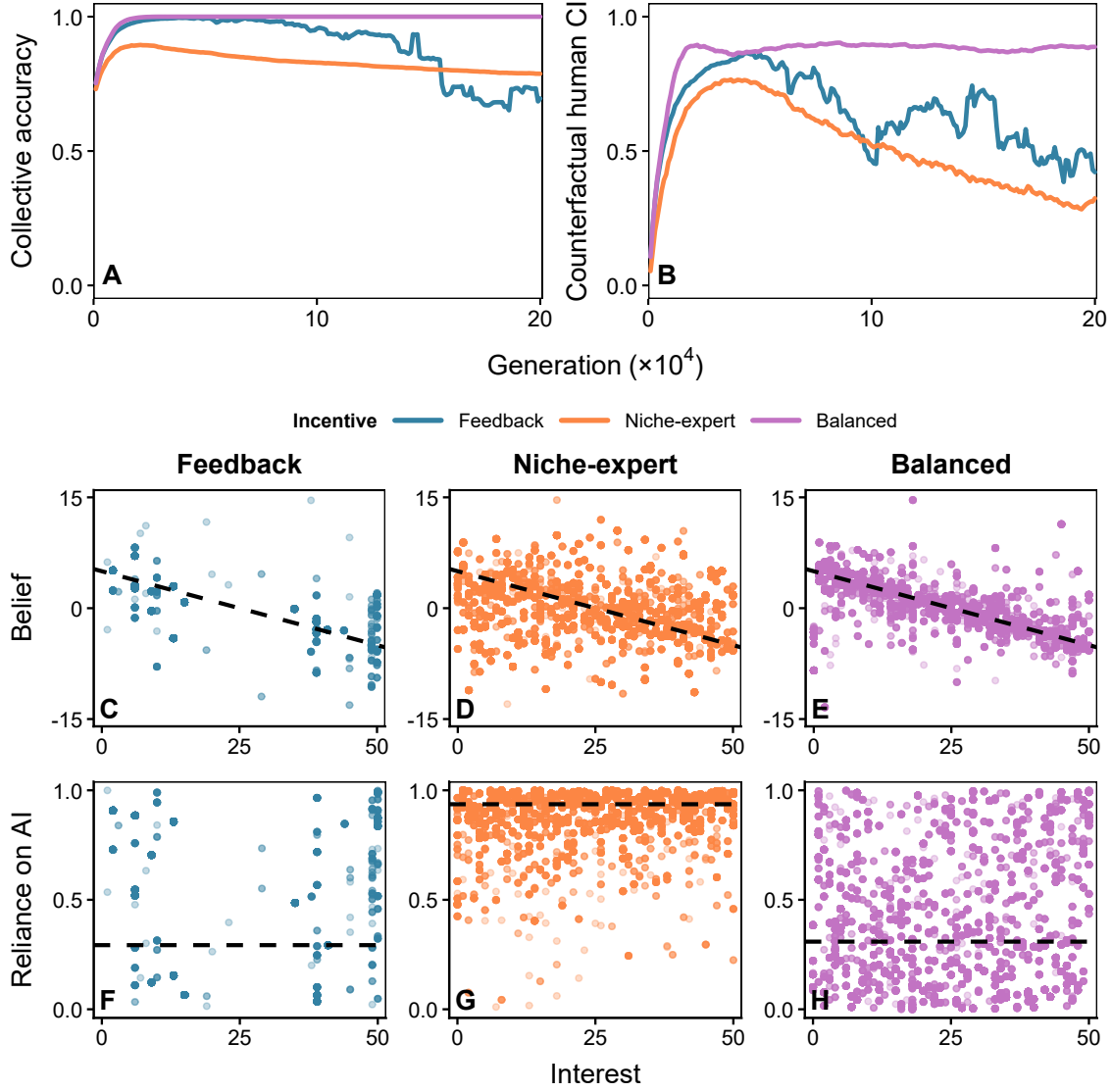


Figure 4: **Balanced structure can promote collective intelligence** Trajectories of collective accuracy (A) and counterfactual human CI (B) under three incentive structures (feedback, niche-expert, and balanced) with Chatbot AI. *[SWP: A and B should have the same y axes scaling.]* Distributions of individual interest and belief (C-E) and of interest and reliance on AI (F-H) after 200,000 generations. Each point represents an individual. Dashed lines in panels C-E represent the true coefficients (α_i) of the prediction target. Dashed horizontal lines in panels F-H indicate the median reliance on AI. Prediction target parameters: $m = 50$, $s = 50$, $\mu = 0$, $\alpha_i = 5 - 0.2i$, and $\sigma_i \sim U(0, 3)$. Population parameters: $N = 10,000$, $c_A \sim \mathcal{N}(0, 5^2)$, and $\beta_A \sim U(0, 1)$. AI parameters: $b_i = 0.4$ and $\tau = 0.3$.

229 a “chatbot” AI model, whose outcome depends on the user’s interest, can impair it.
230 These contrasting outcomes arise because AI assistance affects not only individual
231 predictions, but also the population’s interest diversity, beliefs, and reliance on AI.
232 We also show that directly incentivizing or penalizing AI uses, independent of in-
233 dividual’s performances, can further limit the collective accuracy of the population.
234 Instead, a balanced incentive structure can maintain CI by promoting diversity and
235 limiting bias.

236 Our analysis reveals that AI can prevent the emergence of CI through two dif-
237 ferent mechanisms. First, AI use can cause individuals to concentrate on a subset of
238 the factors that determine the prediction target, thereby reducing diversity in their
239 interests. This result is broadly consistent with recent empirical studies showing that
240 AI assistance can improve individual performance but limit the diversity of outputs,
241 in domains ranging from scientific research [31] to creative writing [17]. Second, over-
242 reliance on AI can cause the population to inherit inherent biases in AI’s predictions,
243 causing the prediction accuracy to decrease eventually. This mechanism can prevent
244 the emergence of CI even when the interest diversity is maintained because biases in
245 AI predictions can be amplified when individual predictions are aggregated. Thus,
246 preserving the diversity of ideas will not be sufficient to promote CI. Instead, AI
247 uses need to be calibrated to prevent systematic AI biases from becoming collective
248 biases.

249 Our analyses of counterfactual collective accuracy also raises an important con-
250 cern for future AI uses: while AI uses may improve the apparent performance of
251 individuals and, in some cases, the population as a whole, it can simultaneously re-
252 duce the population’s ability to make accurate predictions without AI. These results
253 echo findings from a recent field experiment, which demonstrated that students, who
254 seemingly performed better in solving mathematics problems under AI assistance,
255 performed worse on their own than those who did not have access [32]. These findings
256 highlight the need to evaluate AI tools not only by the performance gains they pro-
257 vide during use, but also by whether individuals and populations retain the ability to
258 perform without them. Making this distinction clear will be particularly important
259 for education, where AI may help students complete a task without helping them
260 learn how to do it on their own.

261 As generative AIs and large language models are already generating substantial
262 benefits, many organizations, both industrial and academic, are increasingly seeking
263 to improve their collective performance by regulating AI uses. Such regulations may
264 take direct forms, including AI-use policies or token-based leaderboards, or may op-
265 erate indirectly through other incentives, such as productivity rewards or the social
266 stigma associated with AI use. However, our model suggests that directly incentiviz-
267 ing or penalizing AI use is unlikely to induce appropriately calibrated reliance on
268 AI or to promote CI. Incentivizing AI use could lead to overreliance, causing groups
269 to propagate AI biases, whereas penalizing AI use could lead to underreliance, pre-
270 venting groups from using AI even when it performs well in the relevant domain.
271 Our result instead shows that appropriate AI use and accurate CI can be promoted

by focusing on the traditional properties of groups that have been emphasized for decades: maintaining diversity and preventing collective bias, rather than on AI use itself.

[SWP: need to be edited:] There are several limitations to our analyses. First, the model considers only social learning among humans and treats the influence of AI as unidirectional. In reality, humans can learn from AI, while AI systems can also be further trained on human-generated data. Such bidirectional adaptation may further reduce group performance by increasing the similarity between human and AI predictions and thereby weakening their complementarity. Empirical studies of human–AI collaboration likewise suggest that stronger correlations between human and AI predictions can diminish the benefits of collaboration. Second, we model AI errors using factor-specific additive biases and random error. Previous studies of AI bias and error suggest that additive biases may arise from skewed training data, or systematic calibration. However, AI errors may also exhibit more complex structures, including biased estimates of factor effects and correlations across prediction errors. Further empirical research is therefore needed to characterize the structure of AI bias and error, together with modeling approaches that can incorporate these more complex patterns. Finally, to retain the tractability of established CI models, we assume that the prediction target is a linear combination of causal factors, which does not hold for many real-world tasks. In particular, many machine learning methods used in AI systems are capable of capturing nonlinear relationships. Future research should therefore investigate how nonlinear relationships between causal factors and prediction targets affect the interaction between AI assistance and the emergence of CI.

Methods

Prediction target

Following previous models of CI, we model the prediction target as a linear combination of M independent causal factors:

$$Y = \alpha_0 + \alpha_1 X_1 + \cdots + \alpha_M X_M, \quad (2)$$

where the fixed coefficient α_i determines the magnitude and direction of the effect of factor i , and α_0 is a global intercept. Each causal factor X_i is modeled as an independent Gaussian random variable with mean zero and standard deviation σ_i .

Players

We consider a well-mixed population $\mathcal{N}$ consisting of N players engaged in a novel prediction task. In the absence of AI assistance, the strategy of player A is characterized by two variables: interest e_A and belief c_A . Player A observes the causal

307 factor X_{e_A} and predicts the target by multiplying the observed factor value by their
 308 belief:

$$Y_A = c_A X_{e_A}, \quad (3)$$

309 where e_A can take an integer value from 0 to M . When $e_A = 0$, the player
 310 does not observe any causal factor and instead makes a constant prediction based
 311 solely on their belief. We refer to this as a stubborn strategy because the player's
 312 prediction does not respond to the realized values of the causal factors. For notational
 313 convenience, we assume that $X_0 = 1$.

314 When the group is assisted by AI, each player's strategy additionally includes
 315 reliance on AI, denoted by β_A . We assume that each player consults the AI inde-
 316 pendently and revises their prediction by taking a weighted average of their own
 317 prediction and the AI-generated prediction:

$$Y_A^{\text{re}} = \beta_A Y_A^{\text{AI}} + (1 - \beta_A) Y_A, \quad (4)$$

318 where β_A ranges from 0 to 1. A value of $\beta_A = 1$ indicates complete reliance on
 319 AI, whereas $\beta_A = 0$ indicates that the player does not use the AI prediction.

320 AI models

321 We assume that the AI model has been trained on similar tasks and is therefore
 322 capable of making predictions based on causal factors. However, because the model
 323 is imperfect, its outputs are subject to systematic additive bias and stochastic error.
 324 The additive bias associated with factor i is denoted by b_i , and the stochastic error
 325 is denoted by ϵ . The AI is neither trained nor updated during the simulation, and its
 326 biases therefore remain fixed throughout the simulation. Although the bias is indexed
 327 by factor, all b_i , including b_0 , are assigned the same value in the main analyses for
 328 simplicity. The model formulation nevertheless allows biases to differ among factors.

329 We assume that the stochastic error follows a Gaussian distribution with mean
 330 zero and standard deviation τ . Each time a player consults the AI, an indepen-
 331 dent error is generated. Errors are therefore independent across players and across
 332 repeated consultations by the same player. Consequently, stochastic AI errors can
 333 be attenuated when independently generated predictions are aggregated across play-
 334 ers. We therefore held τ constant and varied the magnitude of the additive bias to
 335 manipulate the level of AI accuracy across analyses.

336 We consider two extreme types of AI that differ in the amount of information
 337 available for prediction. The first is termed the Omniscient AI, which can observe
 338 all M causal factors and incorporate them into its prediction regardless of the user's
 339 interest. The prediction generated by the Omniscient AI for player A is expressed
 340 as follows:

$$Y_A^{\text{AI}} = \alpha_0 + b_0 + \alpha_1 X_1 + b_1 + \cdots + \alpha_M X_M + b_M + \epsilon. \quad (5)$$

341 The second is termed the Chatbot AI, which can use only the single causal factor
 342 X_{e_A} associated with the player’s interest. The prediction generated by the Chatbot
 343 AI for player A is expressed as follows:

$$Y_A^{\text{AI}} = \alpha_{e_A} X_{e_A} + b_{e_A} + \epsilon. \quad (6)$$

344 Aggregation rules

345 The revised individual predictions are aggregated to form a collective prediction. We
 346 use two aggregation rules proposed in previous research, depending on the type of
 347 AI used by the group.

348 When the group is assisted by the Omniscient AI, the AI component of each
 349 revised prediction contains information about all M factors, even though the indi-
 350 vidual’s own prediction is based on only one factor. Each revised prediction can
 351 therefore be treated as an estimate of the entire target. Accordingly, we calculate
 352 the collective prediction by taking the simple average of the revised individual pre-
 353 dictions:

$$\hat{Y}^{\text{re}} = \frac{Y_1^{\text{re}} + Y_2^{\text{re}} + \dots + Y_n^{\text{re}}}{N}. \quad (7)$$

354 When the group is assisted by the Chatbot AI, both the player’s initial prediction
 355 and the AI’s prediction are based on the same causal factor. Each revised predic-
 356 tion therefore contains information about only one factor and can be treated as an
 357 estimate of that factor’s contribution to the target. We consequently use clustering
 358 aggregation.

359 Clustering aggregation is performed by grouping individuals according to their
 360 interests and calculating the mean prediction within each group. The collective
 361 prediction is then obtained by summing the mean predictions of all nonempty groups:

$$\hat{Y}^{\text{re}} = \sum_{i:n_i>0} \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} Y_A^{\text{re}}, \quad (8)$$

362 where $\mathcal{A}_i$ is the set of players interested in factor i , and $n_i = |\mathcal{A}_i|$ is the number
 363 of players in that set.

364 When the group is not assisted by AI, collective predictions are computed from the
 365 unrevised individual predictions throughout the simulation. Both simple averaging
 366 and clustering can be used in the absence of AI. For fair comparison, we use the
 367 same aggregation rule as in the corresponding AI-assisted condition.

368 Incentive structures

369 Players receive payoffs determined by an incentive structure based on their strategies.
 370 We consider five incentive structures: feedback structure, niche-expert structure,
 371 direct intervention on AI use of these two structures, and a balanced structure.

372 The feedback structure does not directly reward individuals according to their
 373 own prediction accuracy. Instead, it rewards predictions that are aligned with the
 374 residual error of the collective prediction and therefore have the potential to move
 375 the collective prediction toward the true value. Given player A 's revised prediction
 376 Y_A^{re} and the collective prediction $\hat{Y}^{\text{re}}$, the payoff is expressed as

$$\pi_A = Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}). \quad (9)$$

377 The niche-expert structure assigns higher payoffs to individuals who make accu-
 378 rate predictions while having an underrepresented interest. Specifically, the squared
 379 prediction error of player A is weighted by $\rho_{e_A} = n_{e_A}/n$, the proportion of players
 380 interested in the same factor:

$$\pi_A = -\rho_{e_A}(Y_A^{\text{re}} - Y)^2. \quad (10)$$

381 Previous research has shown that these two incentive structures can produce
 382 nearly perfect CI through social learning, except when the niche-expert incentive is
 383 combined with simple averaging. They are therefore used as baseline structures for
 384 examining how AI assistance affects the emergence of CI.

385 To examine the effects of directly encouraging or discouraging AI use, we mod-
 386 ify the feedback and niche-expert structures by adding a term proportional to the
 387 player's reliance on AI:

$$\pi_A = Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}) + \lambda\beta_A. \quad (11)$$

$$\pi_A = -\rho_{e_A}(Y_A^{\text{re}} - Y)^2 + \lambda\beta_A. \quad (12)$$

388 Here, λ is a weighting coefficient representing the strength of the direct incentive
 389 or penalty applied to AI use. A positive value of λ rewards AI reliance, whereas a
 390 negative value penalizes it. When $\lambda = 0$, no direct incentive or penalty is applied,
 391 and the payoff functions reduce to the original feedback and niche-expert structures.

392 The balanced structure is defined as the arithmetic mean of the feedback and
 393 niche-expert payoff components:

$$\pi_A = \frac{Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}) - \rho_{e_A}(Y_A^{\text{re}} - Y)^2}{2}. \quad (13)$$

394 Simulations

395 All simulations were conducted using an agent-based model with 50 causal factors
 396 ($M = 50$) and 10,000 players ($n = 10,000$). The parameters defining the prediction
 397 target were sampled at the beginning of the simulations and subsequently held fixed.

398 The coefficients α_i were sampled from a uniform distribution on $[-5, 5]$, except in
 399 the simulation presented in Figure 4, in which α_i was set to $5 - 0.2i$ for visualization.
 400 The standard deviations σ_i were sampled from a uniform distribution on $[0, 3]$.

401 Players’ initial strategies were also sampled from predefined probability distri-
 402 butions. The distribution of individual belief c_A depended on the aggregation rule.
 403 Under simple averaging, each factor’s contribution is scaled by the proportion of
 404 players interested in that factor. We therefore sampled beliefs from a Gaussian dis-
 405 tribution with mean zero and standard deviation 100, following the scaling used
 406 in previous work [13]. Under clustering aggregation, beliefs were sampled from a
 407 Gaussian distribution with mean zero and standard deviation 5. Because predictions
 408 are averaged separately within each interest group and then summed, no additional
 409 scaling of c_A is required. The initial interest of each player was sampled uniformly
 410 from the integers between 0 and 50, and reliance on AI was sampled from a uniform
 411 distribution on $[0, 1]$.

412 We assumed that prediction events occurred on a substantially faster timescale
 413 than social learning. At each generation, we therefore calculated the expected payoff
 414 $\mathbb{E}[\pi_A]$ of each player analytically rather than explicitly sampling prediction events.
 415 The expectation was taken over the distributions of the causal factors $X_1, \dots, X_M$
 416 and the stochastic AI error ϵ , conditional on the current population strategies and
 417 the fixed target parameters. Closed form expression of expected payoffs were derived
 418 for all incentive structures and are presented in the Supplementary Methods.

419 Each generation consisted of an expected payoff calculation followed by one op-
 420 portunity for pairwise social learning. Two distinct players, A and B , were randomly
 421 sampled, and player A imitated all three strategies of player B with probability

$$\frac{1}{1 + \exp(s(\mathbb{E}[\pi_A] - \mathbb{E}[\pi_B]))}, \quad (14)$$

422 where s represents the strength of social learning and was set to 50 in all sim-
 423 ulations. We introduced a mutation rate μ , such that with probability μ , player A
 424 did not imitate player B but instead redrew all strategy components from the same
 425 distributions used to initialize the population. The mutation rate was set to zero in
 426 all simulations except those presented in Supplementary Figure 4.

427 Unless otherwise specified, each simulation was run for 200,000 generations. When
 428 a result was represented by a single summary value rather than as a temporal tra-
 429 jectory, the metric was averaged over the final 10,000 generations.

430 To evaluate the dynamics of collective prediction, we defined six summary met-
 431 rics: collective accuracy, collective bias, collective variance, counterfactual collective
 432 accuracy, interest diversity, and median reliance on AI. At each population state,
 433 expectations and variances were evaluated analytically over the distributions of the
 434 causal factors and AI errors rather than estimated from sampled prediction events.
 435 Collective bias and collective variance constitute the bias–variance decomposition of
 436 the group level collective prediction error.

437 Collective accuracy was normalized relative to the baseline prediction α_0 . A neg-
 438 ative value indicates that the collective prediction performs worse than the baseline.
 439 Counterfactual collective accuracy was evaluated by setting $\beta_A = 0$ while retaining
 440 the evolved interests and beliefs of the AI-assisted population. In calculating interest

441 diversity, we used the convention that $\rho_i \ln \rho_i = 0$ when $\rho_i = 0$ (Table 1).

Table 1: Evaluation metrics used to analyze collective prediction dynamics.

Metric	Definition	Interpretation
Collective accuracy	$1 - \frac{\mathbb{E}[(Y - \hat{Y}^{\text{re}})^2]}{\mathbb{E}[(Y - \alpha_0)^2]}$	Accuracy of the revised collective prediction relative to the baseline prediction α_0 .
Collective bias	$\left(\mathbb{E}[Y - \hat{Y}^{\text{re}}]\right)^2$	Squared expected deviation of the revised collective prediction from the true target.
Collective variance	$\text{Var}(Y - \hat{Y}^{\text{re}})$	Variability of the revised collective prediction error across prediction events.
Counterfactual collective accuracy	$1 - \frac{\mathbb{E}[(Y - \hat{Y})^2]}{\mathbb{E}[(Y - \alpha_0)^2]}$	Accuracy obtained after removing AI use from the current population while retaining its evolved interests and beliefs.
Interest diversity	$\exp\left(-\sum_{i=0}^M \rho_i \ln \rho_i\right)$	Effective number of interests represented in the population.
Median reliance on AI	$\text{median}_{A \in N}(\beta_A)$	Median reliance on AI in the population.

442 Data and code availability

443 All data are stored in a publicly available GitHub repository ([https://github.com/](https://github.com/parklab-snu/collective-intelligence-with-AI)
444 [parklab-snu/collective-intelligence-with-AI](https://github.com/parklab-snu/collective-intelligence-with-AI)).

445 Acknowledgements

446 S.W.P. was supported by the New Faculty Startup Fund from Seoul National Univer-
447 sity, and the Global-LAMP Program of the National Research Foundation of Korea
448 (NRF) grant funded by the Ministry of Education (No. RS-2023-00301976).

449 **Contributions**

450 Jaeseung Kim: Conceptualization, Formal Analysis, Investigation, Methodology,
451 Software, Visualization, Writing - Original Draft Preparation, Writing - Review &
452 Editing. Sang Woo Park: Conceptualization, Investigation, Methodology, Supervi-
453 sion, Writing - Original Draft Preparation, Writing - Review & Editing.

454 **Competing Interests**

455 The authors declare no competing interests.

456 Supplementary Text

457 1 Omniscient AI

458 Omniscient AI assumes that AI can incorporate all causal factors regardless of the
459 player's interest.

460 1.1 Model

$$Y_A^{AI} = \alpha_0 + b_0 + \alpha_1 X_1 + b_1 + \cdots + \alpha_M X_M + b_M + \epsilon. \quad (S1)$$

461 For simplicity, prediction of Omniscient AI can be written as:

$$Y^{AI} = Y + B + \epsilon \quad (S2)$$

462 where Y stands for true prediction target ($Y = \alpha_0 + \alpha_1 X_1 + \cdots + \alpha_M X_M$) and
463 B is sum of all fixed biases ($B = b_0 + b_1 + \cdots + b_M$), ϵ follows normal distribution
464 with zero mean, τ standard deviation with no correlation within or between players.
465 Player A's prediction and revised prediction can be written as follows:

$$Y_A = c_A X_{e_A} \quad (S3)$$

$$Y_A^{re} = \beta_A Y^{AI} + (1 - \beta_A) Y_A \quad (S4)$$

467 where c_A stands for player A's belief on the factor, e_A stands for the interest, and
468 β_A stands for reliance on AI.

469 1.2 Aggregation

470 Because Y_A^{AI} is subject to all M factors, we use simple averaging aggregation for
471 collective prediction.

$$\begin{aligned} \hat{Y}^{re} &= \frac{1}{N} \sum_{j=1}^N Y_j^{re} \\ &= \frac{1}{N} \sum_{j=1}^N \{ \beta_j Y^{AI} + (1 - \beta_j) Y_j \} \\ &= \frac{1}{N} \sum_{j=1}^N \{ \beta_j (Y + B + \epsilon) + (1 - \beta_j) c_j X_{e_j} \} \end{aligned}$$

472 Reparameterization for simple derivation and interpretability.

$$\bar{\beta} := \frac{1}{N} \sum_{j=1}^N \beta_j \quad (S5)$$

473

$$\bar{\eta} := \frac{1}{N} \sum_{j=1}^N \beta_j \epsilon \quad (\text{S6})$$

474

$$\gamma_i := \frac{1}{N} \sum_{A \in \mathcal{A}_i} (1 - \beta_A) c_A \quad (\text{S7})$$

475 $\bar{\beta}$ can be interpreted as mean reliance on AI, γ stands for the coefficient for the factor
 476 i which is solely based on players' belief.

477 By this reparameterization, we can write $\hat{Y}^{re}$ as follows.

$$\hat{Y}^{re} = \bar{\beta}(Y + B) + \sum_{i=0}^M \gamma_i X_i + \bar{\eta} \quad (\text{S8})$$

478 1.3 Expected payoffs

479 Feedback

$$\delta_0 := \alpha_0 - (\bar{\beta}(\alpha_0 + B) + \gamma_0) \quad (\text{S9})$$

480

$$\delta_i := \alpha_i - (\bar{\beta}\alpha_i + \gamma_i), \quad (i \neq 0) \quad (\text{S10})$$

481

$$Y - \hat{Y}^{re} = \delta_0 + \sum_{i=1}^M \delta_i X_i - \bar{\eta} \quad (\text{S11})$$

482 See equation Eq. (S6) for the $\bar{\eta}$.

483 δ_0 is constant term and δ_i is independent of α_k ($k \neq i$), ϵ , and $\bar{\eta}$.

$$\begin{aligned} \mathbb{E}(\pi_A) &= \mathbb{E}[Y_A^{re}(Y - \hat{Y}^{re})] \\ &= \mathbb{E} \left[\left\{ \beta_A \alpha_0 + \beta_A \sum_{i=1}^M \alpha_i X_i + \beta_A B + \beta_A \epsilon + (1 - \beta_A) c_A X_{e_A} \right\} \left\{ \delta_0 + \sum_{i=1}^M \delta_i X_i - \bar{\eta} \right\} \right] \\ &= \beta_A \alpha_0 \delta_0 + \beta_A \sum_{i=1}^M \delta_i \alpha_i \sigma_i^2 + \beta_A B \delta_0 - \frac{1}{N} \beta_A^2 \tau^2 + (1 - \beta_A) c_A \delta_{e_A} \sigma_{e_A}^2 \\ &= \beta_A \left\{ (\alpha_0 + B) \delta_0 + \sum_{i=1}^M \alpha_i \delta_i \sigma_i^2 \right\} + (1 - \beta_A) c_A \delta_{e_A} \sigma_{e_A}^2 - \frac{\beta_A^2 \tau^2}{N} \end{aligned}$$

484 **Niche-expert**

485 If $e_A = 0$, player does not observe causal factor and their unrevised prediction is same
 486 with their belief c_A . Such that, the expected payoff is

$$\begin{aligned}
 \mathbb{E}(\pi_A) &= -\mathbb{E}[\rho_0(Y_A^{re} - Y)^2] \\
 &= -\mathbb{E}[\rho_0\{(\beta_A - 1)Y + \beta_A B + \beta_A \epsilon + (1 - \beta_A)c_A\}^2] \\
 &= -\mathbb{E}\left[\rho_0\left\{(\beta_A - 1)(\alpha_0 - c_A) + \beta_A B + (\beta_A - 1)\sum_{i=1}^M \alpha_i X_i + \beta_A \epsilon\right\}^2\right] \\
 &= -\rho_0\left\{(1 - \beta_A)^2(\alpha_0 - c_A)^2 - 2\beta_A(1 - \beta_A)(\alpha_0 - c_A)B + \beta_A^2 B^2\right. \\
 &\quad \left.+ (1 - \beta_A)^2 \sum_{i \neq 0} \alpha_i^2 \sigma_i^2 + \beta_A^2 \tau^2\right\}
 \end{aligned}$$

487 Else,

$$\begin{aligned}
 \mathbb{E}(\pi_A) &= -\mathbb{E}[\rho_{e_A}(Y_A^{re} - Y)^2] \\
 &= -\mathbb{E}[\rho_{e_A}\{(\beta_A - 1)Y + \beta_A B + \beta_A \epsilon + (1 - \beta_A)c_A X_{e_A}\}^2] \\
 &= -\mathbb{E}\left[\rho_{e_A}\left\{(\beta_A - 1)\alpha_0 + \beta_A B + (\beta_A - 1)(\alpha_{e_A} - c_A)X_{e_A} + (\beta_A - 1)\sum_{i \neq e_A, 0} \alpha_i X_i + \beta_A \epsilon\right\}^2\right] \\
 &= -\rho_{e_A}\left\{(1 - \beta_A)^2 \alpha_0^2 - 2\beta_A(1 - \beta_A)\alpha_0 B + \beta_A^2 B^2 + (1 - \beta_A)^2(\alpha_{e_A} - c_A)^2 \sigma_{e_A}^2\right. \\
 &\quad \left.+ (1 - \beta_A)^2 \sum_{i \neq e_A, 0} \alpha_i^2 \sigma_i^2 + \beta_A^2 \tau^2\right\}
 \end{aligned}$$

488 **Incentivizing or penalizing AI use**

489 Because additive term $\lambda\beta$ is not a random variable, we can simply derive the expected
 490 payoff as

$$\mathbb{E}(\pi_{A, Feedback}) = \mathbb{E}[Y_A^{re}(Y - \hat{Y}^{re}) + \lambda\beta_A] = \mathbb{E}[Y_A^{re}(Y - \hat{Y}^{re})] + \lambda\beta_A. \quad (S12)$$

$$\mathbb{E}(\pi_{A, Niche-expert}) = -\mathbb{E}[\rho_{e_A}(Y_A^{re} - Y)^2 + \lambda\beta_A] = -\mathbb{E}[\rho_{e_A}(Y_A^{re} - Y)^2] + \lambda\beta_A. \quad (S13)$$

491 **1.4 Expected accuracy**

$$\mathbb{E}[(Y - \alpha_0)^2] = \mathbb{E}\left[\left(\alpha_0 + \sum_{i=1}^M \alpha_i X_i - \alpha_0\right)^2\right] = \sum_{i=1}^M \alpha_i^2 \sigma_i^2 \quad (S14)$$

492

$$\mathbb{E}[(Y - \hat{Y}^{re})^2] = \mathbb{E}\left[\left(\delta_0 + \sum_{i=1}^M \delta_i X_i - \hat{\eta}\right)^2\right] = \delta_0^2 + \sum_{i=1}^M \delta_i^2 \sigma_i^2 + \frac{\tau^2}{N^2} \sum_{j=1}^N \beta_j^2 \quad (\text{S15})$$

493

$$\begin{aligned} R &= 1 - \frac{\mathbb{E}[(Y - \hat{Y})^2]}{\mathbb{E}[(Y - \alpha_0)^2]} \\ &= 1 - \frac{\delta_0^2 + \sum_{i=1}^M \delta_i^2 \sigma_i^2 + \frac{\tau^2}{N^2} \sum_{j=1}^N \beta_j^2}{\sum_{i=1}^M \alpha_i^2 \sigma_i^2} \end{aligned}$$

494 2 Chatbot AI

495 2.1 Model

$$Y_A^{AI} = \alpha_{e_A} X_{e_A} + b_{e_A} + \epsilon \quad (\text{S16})$$

496 b_{e_A} stands for factor specific additive bias, ϵ follows normal distribution with zero
 497 mean, τ standard deviation with no correlation within or between players. Player
 498 A's prediction and revised prediction can be written as follows:

$$Y_A = c_A X_{e_A} \quad (\text{S17})$$

499

$$Y_A^{re} = \beta_A Y_A^{AI} + (1 - \beta_A) Y_A \quad (\text{S18})$$

500 where c_A stands for player A's belief on the factor, e_A stands for the interest, and
 501 β_A stands for reliance on AI.

502 2.2 Aggregation

503 For Chatbot AI, clustering aggregation is used for collective prediction.

$$\begin{aligned} \hat{Y}^{re} &= \sum_{i=0}^M \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} Y_A^{re} \\ &= \sum_{i=0}^M \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} \{\beta_A Y_A^{AI} + (1 - \beta_A) Y_A\} \\ &= \sum_{i=0}^M \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} [\{\beta_A \alpha_{e_A} + (1 - \beta_A) c_A\} X_{e_A} + \beta_A b_{e_A} + \beta_A \epsilon] \end{aligned}$$

504 Reparameterize for the simplicity.

$$\tilde{c}_i := \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} \{\beta_A \alpha_{e_A} + (1 - \beta_A) c_A\} \quad (\text{S19})$$

505

$$\tilde{B}_i := \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} \beta_A b_{e_A} \quad (\text{S20})$$

506

$$\bar{B} := \sum_{i=0}^M \tilde{B}_i \quad (\text{S21})$$

507

$$\tilde{\eta}_i := \frac{1}{n_i} \sum_{A \in \mathcal{A}_i} \beta_A \epsilon \quad (\text{S22})$$

508

$$\bar{\eta} := \sum_{i=0}^M \tilde{\eta}_i \quad (\text{S23})$$

509 2.3 Expected payoffs

510 Feedback

511

$$\delta_0 := \alpha_0 - \tilde{c}_0 - \bar{B} \quad (\text{S24})$$

512

$$\delta_i := \alpha_i - \tilde{c}_i, \quad (i \neq 0) \quad (\text{S25})$$

$$Y - \hat{Y}^{re} = \delta_0 + \sum_{i=1}^M \delta_i X_i - \bar{\eta} \quad (\text{S26})$$

513 δ_0 is constant term and δ_i is independent of α_k ($k \neq i$), ϵ , and $\bar{\eta}$.

$$\begin{aligned} \mathbb{E}(\pi_A) &= \mathbb{E}[Y_A^{re}(Y - \hat{Y}^{re})] \\ &= \mathbb{E} \left[\{(\beta_A \alpha_{e_A} + (1 - \beta_A) c_A) X_{e_A} + \beta_A b_{e_A} + \beta_A \epsilon\} \left\{ \delta_0 + \sum_{i=1}^M \delta_i X_i - \bar{\eta} \right\} \right] \\ &= \{\beta_A \alpha_{e_A} + (1 - \beta_A) c_A\} \delta_{e_A} \sigma_{e_A}^2 + \beta_A b_{e_A} \delta_0 - \frac{1}{n_{e_A}} \beta_A^2 \tau^2 \end{aligned}$$

514 Niche-expert

515 If $e_A = 0$,

$$\begin{aligned} \mathbb{E}(\pi_A) &= -\mathbb{E}[\rho_0(Y_A^{re} - Y)^2] \\ &= -\mathbb{E} \left[\rho_0 \left[\{\beta_A \alpha_0 + (1 - \beta_A) c_A\} + \beta_A b_0 + \beta_A \epsilon - \alpha_0 - \sum_{i=1}^M \alpha_i X_i \right]^2 \right] \\ &= -\rho_0 \left\{ (1 - \beta_A)^2 (c_A - \alpha_0)^2 + 2(1 - \beta_A)(c_A - \alpha_0) \beta_A b_0 + \beta_A^2 b_0^2 \right. \\ &\quad \left. + \sum_{i \neq 0} \alpha_i^2 \sigma_i^2 + \beta_A^2 \tau^2 \right\} \end{aligned}$$

516 Else,

$$\begin{aligned}
\mathbb{E}(\pi_A) &= -\mathbb{E}[\rho_{e_A}(Y_A^{\text{re}} - Y)^2] \\
&= -\mathbb{E}\left[\rho_{e_A}\left[\{\beta_A\alpha_{e_A} + (1 - \beta_A)c_A\}X_{e_A} + \beta_A b_{e_A} + \beta_A\epsilon - \alpha_0 - \sum_{i=1}^M \alpha_i X_i\right]^2\right] \\
&= -\rho_{e_A}\left\{(1 - \beta_A)^2(c_A - \alpha_{e_A})^2\sigma_{e_A}^2 + \sum_{i \neq e_A, 0} \alpha_i^2 \sigma_i^2\right. \\
&\quad \left.+ \alpha_0^2 + \beta_A^2 b_{e_A}^2 - 2\alpha_0\beta_A b_{e_A} + \beta_A^2 \tau^2\right\}
\end{aligned}$$

517 Incentivizing or penalizing AI use

518 Because additive term $\lambda\beta$ is not a random variable, we can simply derive the expected
519 payoff as

$$\mathbb{E}(\pi_{A, \text{Feedback}}) = \mathbb{E}\left[Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}) + \lambda\beta_A\right] = \mathbb{E}\left[Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}})\right] + \lambda\beta_A. \quad (\text{S27})$$

$$\mathbb{E}(\pi_{A, \text{Niche-expert}}) = -\mathbb{E}\left[\rho_{e_A}(Y_A^{\text{re}} - Y)^2 + \lambda\beta_A\right] = -\mathbb{E}\left[\rho_{e_A}(Y_A^{\text{re}} - Y)^2\right] + \lambda\beta_A. \quad (\text{S28})$$

520 Balanced

521 Expected payoff of the balanced structure can also be driven by arithmetic mean of
522 expected payoff of feedback and niche-expert.

$$\mathbb{E}(\pi_A) = \mathbb{E}\left[\frac{Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}}) - \rho_{e_A}(Y_A^{\text{re}} - Y)^2}{2}\right] = \frac{\mathbb{E}\left[Y_A^{\text{re}}(Y - \hat{Y}^{\text{re}})\right] - \mathbb{E}\left[\rho_{e_A}(Y_A^{\text{re}} - Y)^2\right]}{2} \quad (\text{S29})$$

523 2.4 Expected accuracy

$$\mathbb{E}[(Y - \alpha_0)^2] = \mathbb{E}\left[\left(\alpha_0 + \sum_{i=1}^M \alpha_i X_i - \alpha_0\right)^2\right] = \sum_{i=1}^M \alpha_i^2 \sigma_i^2 \quad (\text{S30})$$

524

$$\mathbb{E}[(Y - \hat{Y})^2] = \mathbb{E}\left[\left(\delta_0 + \sum_{i=1}^M \delta_i X_i - \hat{\eta}\right)^2\right] = \delta_0^2 + \sum_{i=1}^M \delta_i^2 \sigma_i^2 + \sum_{i=0}^M \frac{\tau^2}{n_i^2} \sum_{A \in \mathcal{A}_i} \beta_A^2 \quad (\text{S31})$$

$$\begin{aligned}
R &= 1 - \frac{\mathbb{E}[(Y - \hat{Y})^2]}{\mathbb{E}[(Y - \alpha_0)^2]} \\
&= 1 - \frac{\delta_0^2 + \sum_{i=1}^M \delta_i^2 \sigma_i^2 + \sum_{i=0}^M \frac{\tau^2}{n_i^2} \sum_{A \in \mathcal{A}_i} \beta_A^2}{\sum_{i=1}^M \alpha_i^2 \sigma_i^2}
\end{aligned}$$

526 3 Evolutionary dynamics

527 For sufficiently large population ($N \rightarrow \infty$), we can approximate the deterministic
528 evolutionary dynamics using the replicator equation. Let λ denote a strategy, p_λ its
529 frequency, and $u_\lambda = \mathbb{E}(\pi_\lambda)$ its expected payoff. The replicator equation is

$$\frac{dp_\lambda}{dt} = p_\lambda(u_\lambda - \bar{u}), \quad (\text{S32})$$

530 where $\bar{u} = \sum_\lambda p_\lambda u_\lambda$. Therefore, at an internal stationary state, all strategies
531 with positive frequency have the same expected payoff. A boundary state can also
532 be stationary because a strategy with exactly zero frequency cannot increase under
533 the replicator equation. Thus, stability of a boundary state needs to be considered
534 separately by examining invasion by a rare strategy.

535 3.1 Omniscient AI

536 3.1.1 Feedback

537 When the population size approaches infinity, the effect of stochastic error on the
538 collective prediction vanishes ($\bar{\eta} \rightarrow 0$) and the effect of stochastic error in the ex-
539 pected payoff, $-\beta^2 \tau^2 / N$, also vanishes. Following prior research [13], we define the
540 Lyapunov function as the expected error of collective prediction

$$V = \mathbb{E} \left[(Y - \hat{Y}^{re})^2 \right]. \quad (\text{S33})$$

541 The function V is non-negative and equals zero when the collective prediction is
542 identical to the target. To examine how V changes under the evolutionary dynamics,
543 we take its time derivative:

$$\begin{aligned}
\frac{dV}{dt} &= -2\mathbb{E} \left[(Y - \hat{Y}^{re}) \frac{d\hat{Y}^{re}}{dt} \right] \\
&= -2\mathbb{E} \left[\left(\delta_0 + \sum_{i=1}^M \delta_i X_i \right) \left\{ \frac{d\bar{\beta}}{dt} (Y + B) + \sum_{i=0}^M \frac{d\gamma_i}{dt} X_i \right\} \right] \\
&= -2 \left[\left\{ (\alpha_0 + B) \delta_0 + \sum_{i=1}^M \alpha_i \delta_i \sigma_i^2 \right\} \frac{d\bar{\beta}}{dt} + \delta_0 \frac{d\gamma_0}{dt} + \sum_{i=1}^M \delta_i \sigma_i^2 \frac{d\gamma_i}{dt} \right].
\end{aligned}$$

544 For notational convenience, we define

$$K = (\alpha_0 + B) \delta_0 + \sum_{i=1}^M \alpha_i \delta_i \sigma_i^2. \quad (\text{S34})$$

545 Although K changes with the population state, it is common to all strategies at a
546 given state and can therefore be treated as a constant with respect to the strategy
547 index λ . We also define

$$G_e = \begin{cases} \delta_0, & e = 0, \\ \delta_e \sigma_e^2, & e \neq 0. \end{cases} \quad (\text{S35})$$

548 By the definitions of $\bar{\beta}$ and γ_i and the replicator equation,

$$\frac{d\bar{\beta}}{dt} = \sum_{\lambda} p_{\lambda} (u_{\lambda} - \bar{u}) \beta_{\lambda} = \text{Cov}(\beta, u). \quad (\text{S36})$$

549 Similarly, the terms involving γ_i can be combined as

$$\delta_0 \frac{d\gamma_0}{dt} + \sum_{i=1}^M \delta_i \sigma_i^2 \frac{d\gamma_i}{dt} = \text{Cov}((1 - \beta) c G_e, u). \quad (\text{S37})$$

550 Therefore,

$$\begin{aligned}
\frac{dV}{dt} &= -2 \{ K \text{Cov}(\beta, u) + \text{Cov}((1 - \beta) c G_e, u) \} \\
&= -2 \text{Cov}(K\beta + (1 - \beta) c G_e, u).
\end{aligned}$$

551 For the feedback structure in the limit $N \rightarrow \infty$, the expected payoff of strategy
552 λ is

$$u_{\lambda} = K \beta_{\lambda} + (1 - \beta_{\lambda}) c_{\lambda} G_{e_{\lambda}}. \quad (\text{S38})$$

553 Which is same with Thus,

$$\frac{dV}{dt} = -2 \text{Cov}(u, u) = -2 \text{Var}(u) \leq 0. \quad (\text{S39})$$

Therefore, the expected error of collective prediction cannot increase during the evolutionary process and decreases whenever strategies differ in their expected payoffs. The derivative becomes zero when the variance of expected payoffs is zero, $\text{Var}(u) = 0$. This condition corresponds to equal expected payoffs among the strategies present in the population. A boundary state in which the entire population consists of a single strategy can also satisfy $\text{Var}(u) = 0$ even when $V > 0$. However, such a state is not evolutionarily stable under the feedback structure. If the collective prediction differs from the target Y , a nonzero residual remains. By the definition of the feedback payoff, a novel strategy whose prediction aligns with the direction of this residual obtains a higher expected payoff than the resident strategy and can therefore invade the population. Thus, an imperfect boundary equilibrium with $V > 0$ is unstable to the introduction of novel predictions.

Consequently, the evolutionary dynamics continue to reduce the collective prediction error until the population reaches the stable state with zero residual, $V = 0$. At this state, the collective prediction equals the target, $\hat{Y}^{re} = Y$, and the collective accuracy reaches $R = 1$.

3.1.2 Niche-expert

Unlike the stochastic error in the collective prediction, stochastic error in the niche-expert payoff is evaluated at the individual level and does not vanish as population size increases. Therefore, we retain $\tau > 0$ in the following derivation.

We first consider strategies having the same interest by optimizing the expected payoff with respect to c and β . For a nonzero interest e , the expected payoff is strictly concave with respect to c for $\beta < 1$ and $\sigma_e > 0$, and is maximized at

$$c_e^* = \alpha_e. \quad (\text{S40})$$

After substituting $c_e^* = \alpha_e$, the payoff is strictly concave with respect to β . Its unconstrained stationary point is

$$\tilde{\beta}_e = \frac{\alpha_0^2 + \alpha_0 B + \sum_{j \neq e, 0} \alpha_j^2 \sigma_j^2}{(\alpha_0 + B)^2 + \sum_{j \neq e, 0} \alpha_j^2 \sigma_j^2 + \tau^2}. \quad (\text{S41})$$

Because β is constrained between 0 and 1, the optimal β_e^* is $\tilde{\beta}_e$ when $0 < \tilde{\beta}_e < 1$, and otherwise lies at the corresponding boundary, $\beta_e^* = 0$ or $\beta_e^* = 1$. Thus, the boundary cases do not require a separate evolutionary argument; they only change the optimal prediction and its minimum expected squared error. When $\beta_e^* = 1$, the revised prediction is independent of c_e , so different values of c_e are selectively neutral and can be treated as a single payoff-equivalent strategy class.

Let L_e denote the minimum expected individual error, $\mathbb{E}[(Y_A^{re} - Y)^2]$, within interest e . For an internal optimum,

$$L_e = \frac{B^2 \sum_{j \neq e, 0} \alpha_j^2 \sigma_j^2 + \tau^2 \left(\alpha_0^2 + \sum_{j \neq e, 0} \alpha_j^2 \sigma_j^2 \right)}{(\alpha_0 + B)^2 + \sum_{j \neq e, 0} \alpha_j^2 \sigma_j^2 + \tau^2}. \quad (\text{S42})$$

587 If the optimum is located at $\beta_e^* = 0$ or $\beta_e^* = 1$, L_e is instead obtained by evaluating
 588 the expected squared prediction error at the corresponding boundary.

589 For interest 0, optimization with respect to c gives

$$c_0^*(\beta) = \alpha_0 - \frac{\beta}{1 - \beta} B. \quad (\text{S43})$$

590 Substituting this into the expected payoff gives a strictly concave function of β ,
 591 with the unique optimum

$$\beta_0^* = \frac{\sum_{j=1}^M \alpha_j^2 \sigma_j^2}{\sum_{j=1}^M \alpha_j^2 \sigma_j^2 + \tau^2}. \quad (\text{S44})$$

592 Therefore,

$$c_0^* = \alpha_0 - \frac{B}{\tau^2} \sum_{j=1}^M \alpha_j^2 \sigma_j^2, \quad (\text{S45})$$

593 and

$$L_0 = \frac{\tau^2 \sum_{j=1}^M \alpha_j^2 \sigma_j^2}{\sum_{j=1}^M \alpha_j^2 \sigma_j^2 + \tau^2}. \quad (\text{S46})$$

594 We denote the optimal strategy, or the payoff-equivalent strategy class when
 595 $\beta_e^* = 1$, by S_e . A state containing a suboptimal strategy within an interest can
 596 be stationary if the optimal strategy is absent, but it is unstable against invasion
 597 because the optimal strategy has a strictly higher payoff at the same ρ_e . Therefore,
 598 a stable state can contain only S_e within each represented interest.

599 We next consider the frequencies of the $M + 1$ optimal strategies. Let ρ_e denote
 600 the frequency of S_e . Its expected payoff is

$$u_e = -\rho_e L_e. \quad (\text{S47})$$

601 Assuming $L_e > 0$ for all interests, the equal-payoff condition of the replicator
 602 equation at an internal stationary state gives

$$\rho_0^* L_0 = \rho_1^* L_1 = \cdots = \rho_M^* L_M. \quad (\text{S48})$$

603 Together with $\sum_{e=0}^M \rho_e^* = 1$, this uniquely determines

$$\rho_e^* = \frac{D}{L_e}, \quad D = \left(\sum_{j=0}^M \frac{1}{L_j} \right)^{-1}. \quad (\text{S49})$$

604 This internal stationary state is locally stable. At equilibrium, all represented
 605 interests have the same expected payoff, with $\rho_e^* L_e = D$. If the frequency of one
 606 interest increases slightly above its equilibrium value, its payoff decreases because
 607 the niche-expert payoff is proportional to $-\rho_e L_e$. Conversely, if the frequency of
 608 an interest decreases below its equilibrium value, its payoff increases. The replicator
 609 dynamics therefore generate a restoring force that reduces small deviations in interest
 610 frequencies and drives the population back toward the internal stationary state.

611 If one or more interests are absent, the resulting boundary state can be stationary
 612 under the replicator equation, but it is unstable against invasion. The payoff of a rare
 613 absent interest approaches zero as $\rho_e \rightarrow 0$, whereas the mean payoff of the resident
 614 population is negative when $L_e > 0$, so the rare interest increases when introduced
 615 at sufficiently small frequency.

616 Thus, there exists a unique stable full-support equilibrium in terms of the optimal
 617 prediction strategies and their frequencies. Whether the constrained optimum has
 618 $\beta_e^* = 0$, $0 < \beta_e^* < 1$, or $\beta_e^* = 1$ does not change the frequency argument, because
 619 these cases differ only through the resulting value of L_e . When $\beta_e^* = 1$, the strategy
 620 is selectively neutral with respect to c_e , but its prediction, payoff, and equilibrium
 621 interest frequency remain uniquely determined.

622 This equilibrium also allows a direct comparison with the expected accuracy of
 623 the AI. Using Y^{AI} itself without aggregation as the prediction gives

$$\mathbb{E}[(Y - Y^{\text{AI}})^2] = B^2 + \tau^2. \quad (\text{S50})$$

624 For every interest, complete reliance on AI, $\beta = 1$, is a feasible strategy. There-
 625 fore, because L_e is the minimum individual expected squared prediction error within
 626 interest e ,

$$L_e \leq B^2 + \tau^2. \quad (\text{S51})$$

627 Moreover, for interest 0,

$$L_0 = \frac{\tau^2 \sum_{j=1}^M \alpha_j^2 \sigma_j^2}{\sum_{j=1}^M \alpha_j^2 \sigma_j^2 + \tau^2} < \tau^2 \leq B^2 + \tau^2. \quad (\text{S52})$$

628 Because the stable equilibrium has $\rho_0^* > 0$,

$$\sum_{e=0}^M \rho_e^* L_e < B^2 + \tau^2. \quad (\text{S53})$$

629 The collective prediction is the average of individual revised predictions. By
 630 convexity of the squared error,

$$\mathbb{E}[(Y - \hat{Y}^{re})^2] \leq \sum_{e=0}^M \rho_e^* L_e. \quad (\text{S54})$$

631 Therefore,

$$\mathbb{E}[(Y - \hat{Y}^{re})^2] < \mathbb{E}[(Y - Y^{AI})^2]. \quad (\text{S55})$$

632 Because the collective accuracy and AI accuracy use the same normalization by
 633 $\mathbb{E}[(Y - \alpha_0)^2]$, the stationary collective accuracy under the niche-expert structure is
 634 strictly greater than the expected accuracy of the AI prediction.

635 Although the niche-expert structure improves collective prediction relative to the
 636 AI, the stationary collective accuracy does not reach one when $\tau > 0$. For a nonzero
 637 factor i , the residual coefficient of the stationary collective prediction can be written
 638 as

$$\delta_i^* = \alpha_i \{1 - \bar{\beta}^* - \rho_i^*(1 - \beta_i^*)\} = \alpha_i \sum_{e \neq i} \rho_e^*(1 - \beta_e^*). \quad (\text{S56})$$

639 At the stable equilibrium, $\rho_0^* > 0$. In addition, $\tau > 0$ implies $\beta_0^* < 1$. Therefore,
 640 for every factor with $\alpha_i \neq 0$,

$$\delta_i^* \neq 0. \quad (\text{S57})$$

641 Because $\sum_{i=1}^M \alpha_i^2 \sigma_i^2 > 0$, at least one factor contributes positive prediction error.
 642 Hence,

$$\mathbb{E}[(Y - \hat{Y}^{re})^2] > 0, \quad (\text{S58})$$

643 and consequently the stationary collective accuracy satisfies $R^* < 1$.

644 3.2 Chatbot AI

645 3.2.1 Feedback

646 With Chatbot AI, the effect of stochastic error on the collective prediction depends
 647 on n_i , requiring the additional assumption that $n_i \rightarrow \infty$ for every interest for the
 648 stochastic error to vanish. However, Under this assumption, the loss of interest
 649 diversity cannot occur, and therefore the stochastic breakdown of collective accuracy
 650 observed in finite populations cannot be analytically captured. Instead, by assuming
 651 that each interest is represented by infinitely many players, we can show that the
 652 expected collective prediction error monotonically decreases until perfect accuracy
 653 is achieved. Because finite population dynamics are more relevant to the empirical
 654 populations considered in this study, we retain finite population simulations for the
 655 main result.

656 3.2.2 Niche-expert

657 We first consider strategies having the same interest by optimizing the expected
 658 payoff with respect to c and β . For a nonzero interest e , the expected payoff is
 659 strictly concave with respect to c for $\beta < 1$ and $\sigma_e > 0$, and is maximized at

$$c_e^* = \alpha_e. \quad (\text{S59})$$

660 After substituting $c_e^* = \alpha_e$, the payoff is strictly concave with respect to β . Its
 661 unconstrained stationary point is

$$\tilde{\beta}_e = \frac{2\alpha_0 b_e}{2b_e^2 + 2\tau^2}. \quad (\text{S60})$$

662 Because β is constrained between 0 and 1, the optimal β_e^* is $\tilde{\beta}_e$ when $0 < \tilde{\beta}_e < 1$,
 663 and otherwise lies at the corresponding boundary, $\beta_e^* = 0$ or $\beta_e^* = 1$. Because we
 664 assumes AI model with moderate accuracy, b_e and τ is relatively smaller than α_0
 665 which makes $\tilde{\beta}_e > 1$ when $\alpha_0 b_e > 0$ and $\tilde{\beta}_e < 0$ when $\alpha_0 b_e < 0$. These boundary
 666 cases lead to extreme reliance on AI or complete exclusion of AI which both hinders
 667 the emergence of CI. Only when the prediction task is centered near zero and AI is
 668 relatively inaccurate ($\alpha_0 b_e < b_e^2 + \tau^2$), and $\alpha_0 b_e > 0$, the internal optimum β_e^* can be
 669 exist.

670 For interest 0, optimization with respect to c gives

$$c_0^*(\beta) = \alpha_0 - \frac{\beta}{1 - \beta} b_0. \quad (\text{S61})$$

671 Substituting this into the expected payoff gives a strictly concave function of β , with
 672 the unique optimum

$$\beta_0^* = 0. \quad (\text{S62})$$

673 **Supplementary Figure**

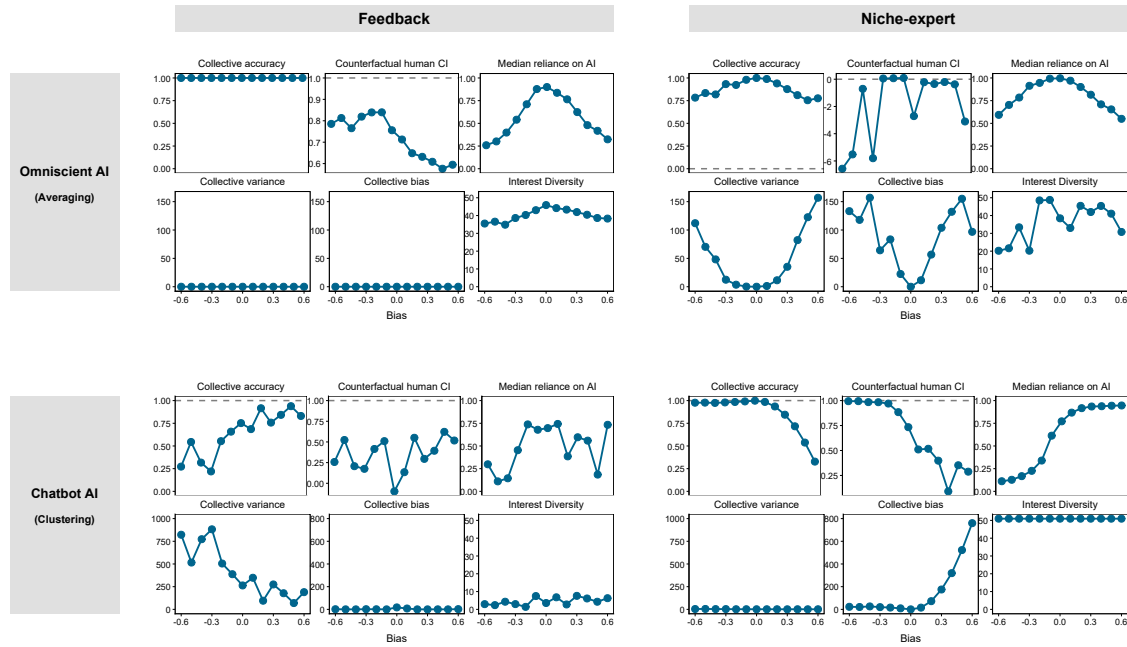


Figure S1: **Caption.** Text.

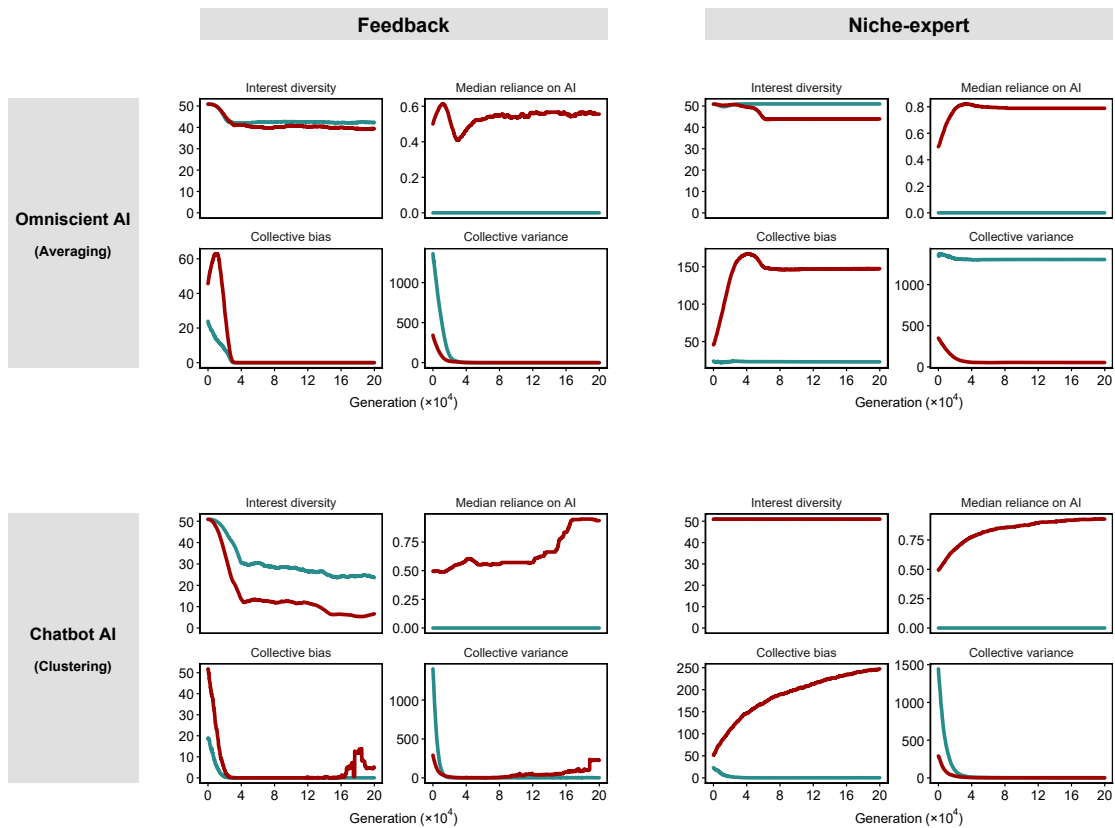


Figure S2: **Caption.** Text.

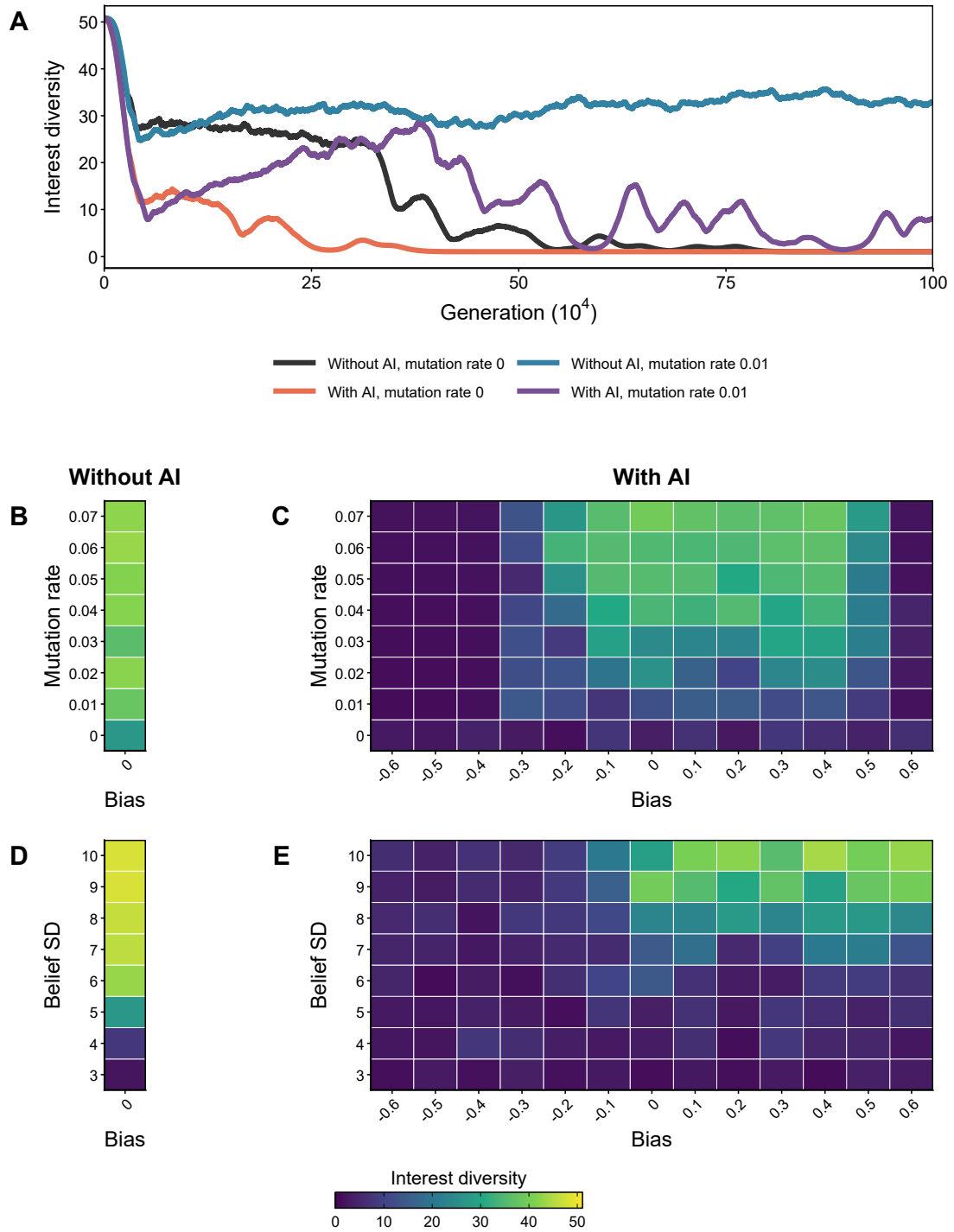


Figure S3: **Caption.** Text.

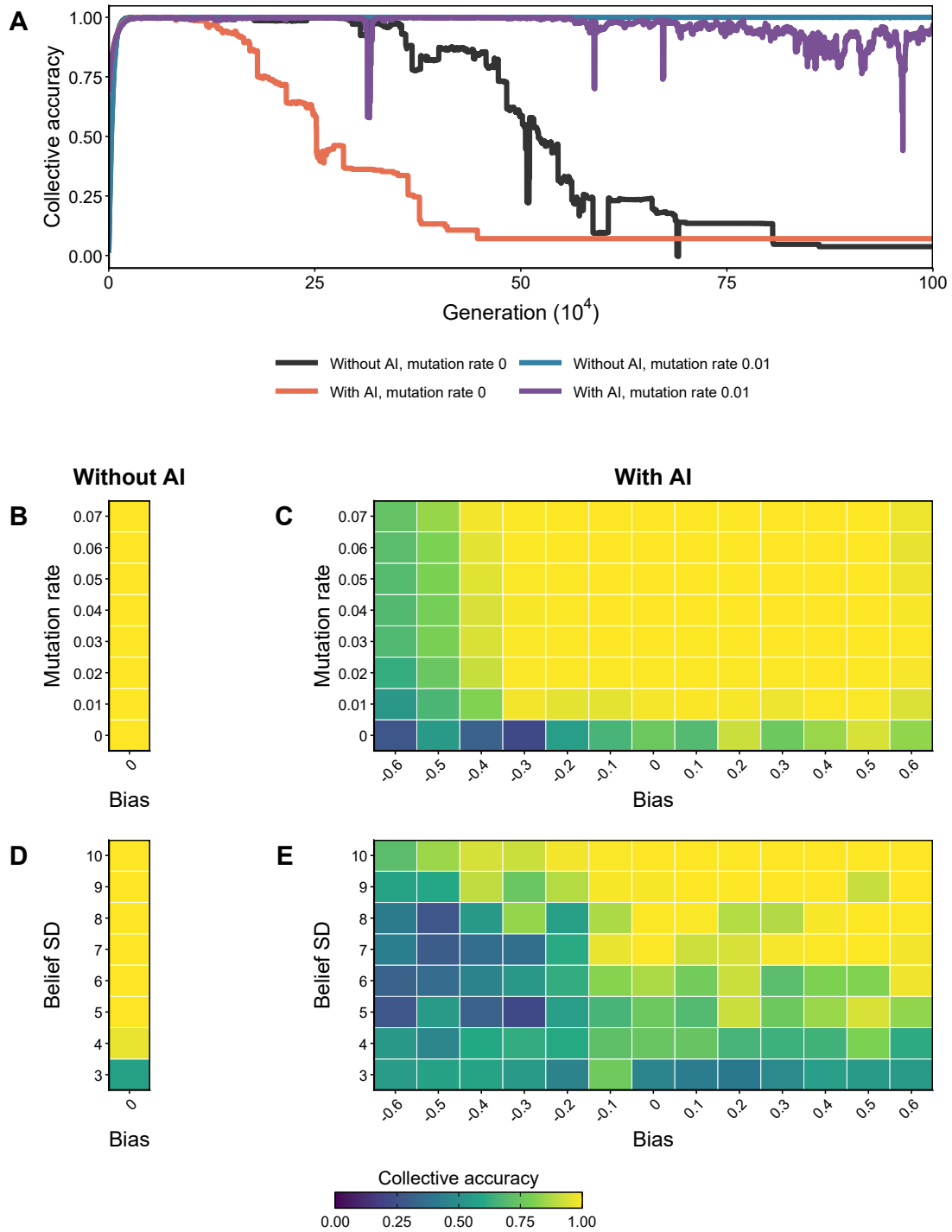


Figure S4: **Caption.** Text.

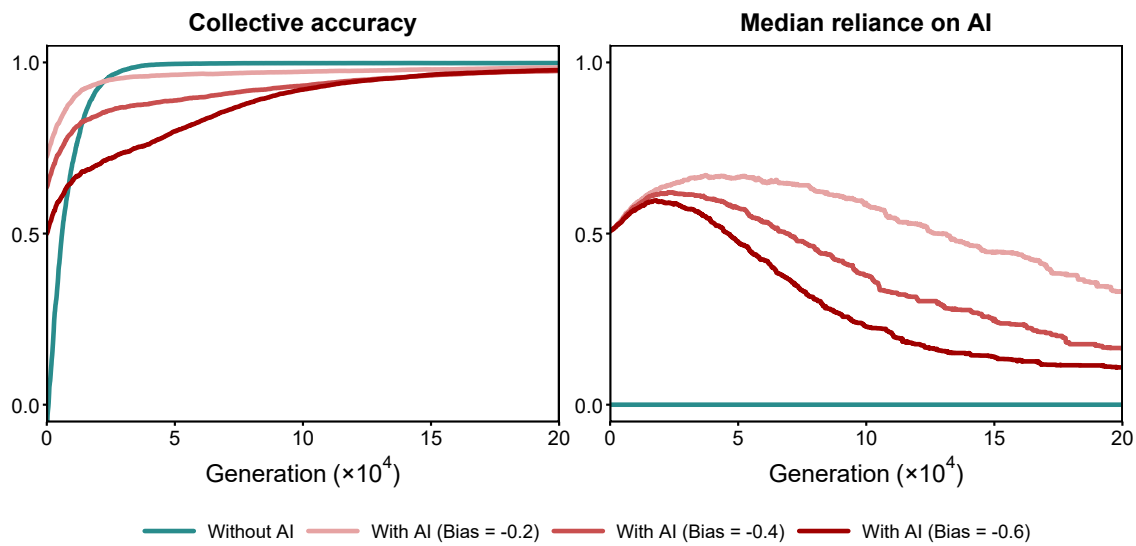


Figure S5: **Caption.** Text.

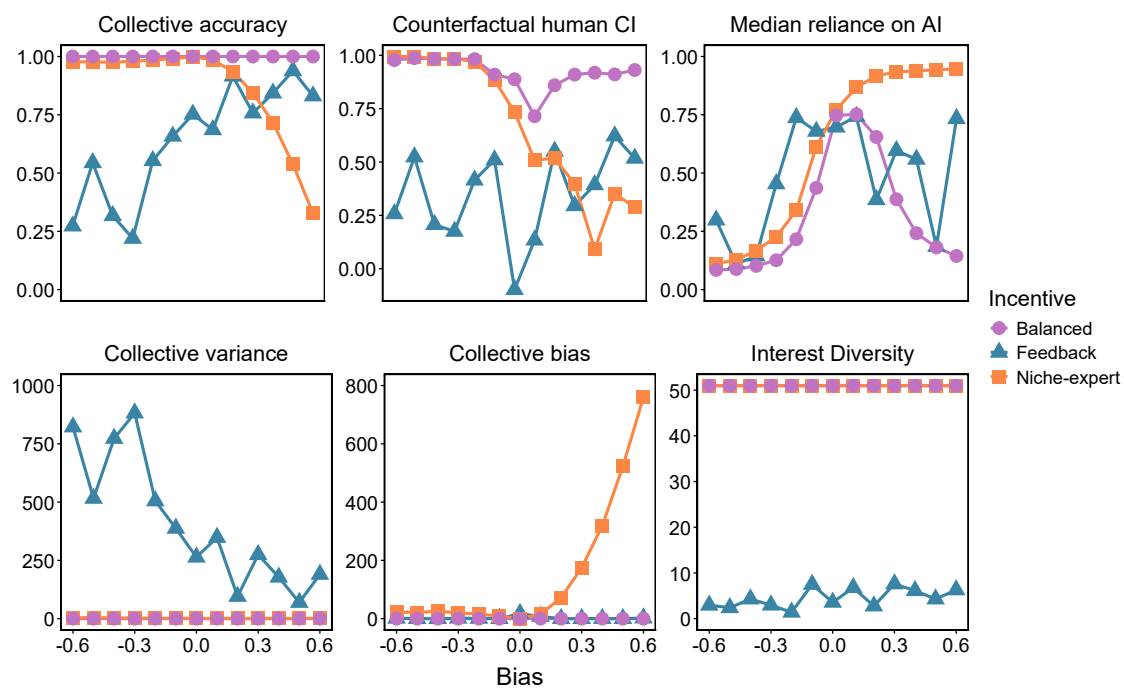


Figure S6: **Caption.** Text.

References

- [1] Francis Galton. Vox populi, 1907.
- [2] Anita Williams Woolley, Christopher F Chabris, Alex Pentland, Nada Hashmi, and Thomas W Malone. Evidence for a collective intelligence factor in the performance of human groups. *science*, 330(6004):686–688, 2010.
- [3] Ilan Yaniv and Maxim Milyavsky. Using advice from multiple sources to revise and improve judgments. *Organizational Behavior and Human Decision Processes*, 103(1):104–120, 2007.
- [4] Cars Hommes, Joep Sonnemans, Jan Tuinstra, and Henk Van de Velden. Coordination of expectations in asset pricing experiments. *The Review of Financial Studies*, 18(3):955–980, 2005.
- [5] Irving Lorge, David Fox, Joel Davitz, and Marlin Brenner. A survey of studies contrasting the quality of group performance and individual performance, 1920–1957. *Psychological bulletin*, 55(6):337, 1958.
- [6] Tabea Dreyer, Amir Haluts, Amos Korman, Nir Gov, Ehud Fonio, and Ofer Feinerman. Comparing cooperative geometric puzzle solving in ants versus humans. *Proceedings of the National Academy of Sciences*, 122(1):e2414274121, 2025.
- [7] Andrew M Simons. Many wrongs: the advantage of group navigation. *Trends in ecology & evolution*, 19(9):453–455, 2004.
- [8] Andrew Berdahl, Colin J Torney, Christos C Ioannou, Jolyon J Faria, and Iain D Couzin. Emergent sensing of complex environments by mobile animal groups. *Science*, 339(6119):574–576, 2013.
- [9] Christoph Riedl, Young Ji Kim, Pranav Gupta, Thomas W Malone, and Anita Williams Woolley. Quantifying collective intelligence in human groups. *Proceedings of the National Academy of Sciences*, 118(21):e2005737118, 2021.
- [10] Leo Breiman. Bagging predictors. *Machine learning*, 24(2):123–140, 1996.
- [11] Leo Breiman. Random forests. *Machine learning*, 45(1):5–32, 2001.
- [12] Richard P Mann and Dirk Helbing. Optimal incentives for collective intelligence. *Proceedings of the National Academy of Sciences*, 114(20):5077–5082, 2017.
- [13] Guocheng Wang, Qi Su, Long Wang, and Joshua B Plotkin. Individual incentives that promote collective intelligence. *Proceedings of the National Academy of Sciences*, 122(51):e2516535122, 2025.

- [14] Justin Werfel. Fluid thinking about collective intelligence. *Nature Machine Intelligence*, 8(4):506–516, 2026.
- [15] Hao Cui and Taha Yasseri. AI-enhanced collective intelligence. *Patterns*, 5(11):101074, 2024.
- [16] Chris Lu, Cong Lu, Robert Tjarko Lange, Yutaro Yamada, Shengran Hu, Jakob Foerster, David Ha, and Jeff Clune. Towards end-to-end automation of AI research. *Nature*, 651(8107):914–919, 2026.
- [17] Anil R Doshi and Oliver P Hauser. Generative AI enhances individual creativity but reduces the collective diversity of novel content. *Science advances*, 10(28):eadn5290, 2024.
- [18] Sha Sajadieh, Loredana Fattorini, Raymond Perrault, Yolanda Gil, Vanessa Parli, Lapo Santarasci, Juan Pava, Nestor Maslej, Russ Altman, Erik Brynjolfsson, et al. Artificial intelligence index report 2026. *arXiv preprint arXiv:2606.15708*, 2026.
- [19] Michelle Vaccaro, Abdullah Almaatouq, and Thomas Malone. When combinations of humans and AI are useful: A systematic review and meta-analysis. *Nature Human Behaviour*, 8(12):2293–2303, 2024.
- [20] Jason W Burton, Ezequiel Lopez-Lopez, Shahar Hechtlinger, Zoe Rahwan, Samuel Aeschbach, Michiel A Bakker, Joshua A Becker, Aleks Berditchevskaia, Julian Berger, Levin Brinkmann, et al. How large language models can reshape collective intelligence. *Nature human behaviour*, 8(9):1643–1655, 2024.
- [21] Lennart Meincke, Gideon Nave, and Christian Terwiesch. ChatGPT decreases idea diversity in brainstorming. *Nature human behaviour*, 9(6):1107–1109, 2025.
- [22] Andreas Fügener, Jörn Grahl, Alok Gupta, and Wolfgang Ketter. Will humans-in-the-loop become borgs? merits and pitfalls of working with AI. *MIS quarterly*, 45(3):1527–1556, 2021.
- [23] Jan Lorenz, Heiko Rauhut, Frank Schweitzer, and Dirk Helbing. How social influence can undermine the wisdom of crowd effect. *Proceedings of the national academy of sciences*, 108(22):9020–9025, 2011.
- [24] Lucy M Aplin, Damien R Farine, Richard P Mann, and Ben C Sheldon. Individual-level personality influences social foraging and collective behaviour in wild birds. *Proceedings of the Royal Society B: Biological Sciences*, 281(1789):20141016, 2014.
- [25] Lu Hong and Scott E Page. Groups of diverse problem solvers can outperform groups of high-ability problem solvers. *Proceedings of the National Academy of Sciences*, 101(46):16385–16389, 2004.

- 743 [26] Isabelle Bousquette. Why some companies say AI ‘token-
744 maxx’ is key to survival. [https://www.wsj.com/cio-journal/
745 why-some-companies-say-ai-tokenmaxxing-is-key-to-survival-e699a128](https://www.wsj.com/cio-journal/why-some-companies-say-ai-tokenmaxxing-is-key-to-survival-e699a128),
746 April 2026. The Wall Street Journal. Accessed: 2026-07-07.
- 747 [27] Tim Keary. Is the cult of ‘tokenmaxxing’ just another fad or the
748 new normal? [https://www.forbes.com/sites/timkeary/2026/04/13/
749 is-the-cult-of-tokenmaxxingjust-another-fad-or-the-new-normal/](https://www.forbes.com/sites/timkeary/2026/04/13/is-the-cult-of-tokenmaxxingjust-another-fad-or-the-new-normal/),
750 April 2026. Forbes. Accessed: 2026-07-07.
- 751 [28] Dalmeet Singh Chawla. Researchers who use hallucinated references to face
752 arXiv ban. *Nature*, 653(8116):988–989, 2026.
- 753 [29] H Holden Thorp. Chatgpt is fun, but not an author. *Science*, 379(6630):313–313,
754 2023.
- 755 [30] Lu Hong, Scott E Page, and Maria Riolo. Incentives, information, and emergent
756 collective accuracy. *Managerial and Decision Economics*, 33(5-6):323–334, 2012.
- 757 [31] Qian Yue Hao, Fengli Xu, Yong Li, and James Evans. Artificial intelligence tools
758 expand scientists’ impact but contract science’s focus. *Nature*, 649(8099):1237–
759 1243, 2026.
- 760 [32] Hamsa Bastani, Osbert Bastani, Alp Sungu, Haosen Ge, Özge Kabakcı, and
761 Rei Mariman. Generative AI without guardrails can harm learning: Evidence
762 from high school mathematics. *Proceedings of the National Academy of Sciences*,
763 122(26):e2422633122, 2025.